ASSIGNMENT 2:
Lifting-line theory for planar wings

Assignment format

A PDF file report of at max **1500 words** (3 pages of text, see how to find the word count in Overleaf or Word). There is otherwise not a page limit, but you should *only* include content that you deem relevant to explain your findings. You should also fulfil the requirements specified below.
For each group, only **one** person should upload the final report.
The **deadline** for the report can be found on DTU Learn.

Report Objectives

The purpose of this assignment is to use lifting-line theory to determine the aerodynamics properties of planar wings having different plan forms. As basis for the assignment, consider everywhere in the following a planar wing consisting of NACA 2410 airfoil sections operating at a Reynolds number, $Re = 5 \cdot 10^6$.

In the report you are expected to give the necessary information for explaining what you did and to discuss the validity of the obtained results. You should not derive equations but show those that are important for understanding what you are doing. Remember to attach the MatLab or Python script in the appendix when uploading the report.

Fig. 1. The British Spitfire is an example of an aircraft using an elliptic wing plan form.

The report should accomplish the following:

1. Compute the wing lift coefficient, C_L, the induced drag coefficient, $C_{D,i}$, and the friction drag coefficient, C_D, for an elliptic wing with aspect ratios AR = 4, 6, 8, 10, and ∞.
   Present the results in plots showing each aerodynamic variable as a function of angles of attack, with *AoA* running from -4° to 8°. Use one plot for each aerodynamic variable, showing comparative curves for the different aspect ratios.

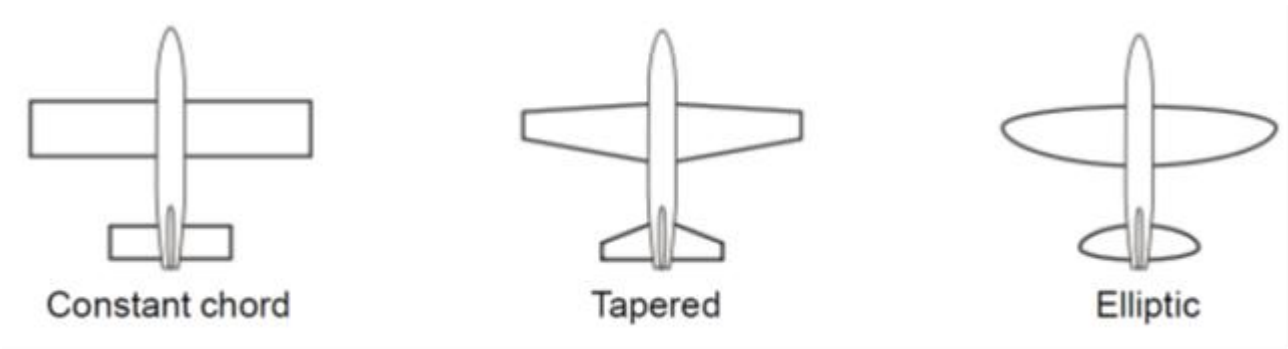

Fig. 2. Wings of different plan forms, Left: Rectangular; Middle: Tapered; Right: Elliptic.

2. Employ the lifting-line theory to determine the distribution of induced angle of attack, α_i, along a rectangular wing (constant chord) with the same aspect ratios, *AR*, as for the elliptic wing in the first question. Depending on what you find most convenient, you may here choose to use either the original formulation of Prandtl or the one by Glauert.
   a. Plot the α_i-distributions along the wingspan for angles of attack (α) 0°, 4°, and 8°. For clarity, plot the results in three figures, one for each *AoA*, with five curves in each (corresponding to the five aspect ratios).
   b. Compute the corresponding lift and induced drag coefficients for the wing, C_L and C_D. Present the results in plots showing each of the two aerodynamic variables as function of angles of attack, with *AoA* running from -4° to 8°. Like in question 1., use one plot for each aerodynamic variable, displaying comparative curves for the different aspect ratios.
3. Consider now a tapered wing of aspect ratio *AR* = 6, at an angle of attack, *AoA* = 5°, and taper ratios *TR* = 0.2, 0.4, 0.6, 0.8 and 1.0. The taper ratio is defined as the ratio between the length of the tip chord and the root (mid) chord, $TR = \frac{c_{tip}}{c_{root}}$.
   a. Introducing the mean chord length, $\bar{c}$, and the dimensionless span-wise coordinate, $\tilde{x} = \frac{x}{\frac{1}{2}b}$, where $x \in \left[-\frac{b}{2}, \frac{b}{2}\right]$. Derive an analytical expression for the dimensionless chord distribution, $\frac{c(\tilde{x})}{\bar{c}}$ as a function of $\tilde{x}$-coordinate, taper and aspect ratio.
   b. Compute and plot the distributions of dimensionless circulation, $\tilde{\Gamma} = \frac{\Gamma}{\bar{c}U_\infty}$, where U_∞ denotes the inflow velocity, the induced angle of attack α_i [deg], and the corresponding lift and induced drag coefficients, C_L and C_D, as functions of the spanwise coordinate, $\tilde{x}$.

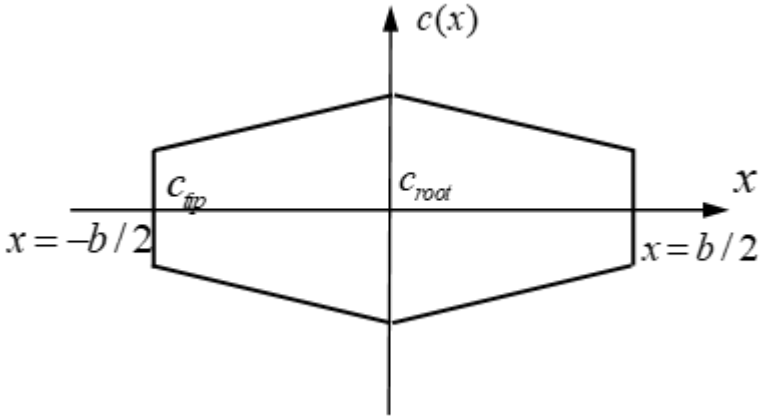

Fig. 3. Sketch showing coordinate system of tapered wing plan form

4. Consider now a <u>rectangular and twisted wing</u> of aspect ratio *AR* = 6, where the wing is twisted with a linear twist from the midpoint of the wing (the root) to the tip. With a linear twist, the equation for the local geometric angle of attack is given as $\alpha_{geom}(x) = \alpha_{g,0} + (\alpha_{g,tip} - \alpha_{g,0}) \cdot |\tilde{x}|$, where $\alpha_{g,0}$ is the geometric angle at the midpoint of the wing and $\alpha_{g,tip}$ is the geometric angle at the tip. Assume now $\alpha_{g,0}$= 2°, and let the angles at the tip take the values $\alpha_{g,tip}$= [0°, 2°, 4°, 6°, 8°].
   For each tip angle, compute and plot the distributions of dimensionless circulation, $\tilde{\Gamma}$, the induced angle of attack α_i, and the corresponding induced drag and lift coefficients, C_l and C_d, as functions of dimensionless span-wise coordinate. Show the results on four figures (one for each variable) with five plots on each (one for each tip angle).

Required figures and tables

- *Question 1:* Plot C_L, α_i, $C_{D,i}$ and C_D, for AR = 4, 6, 8, 10, and ∞. Present the results in plots showing each aerodynamic variable as a function of angles of attack, with *AoA* running from -4° to 8°.
- *Question 2*:
  - Plot of α_i-distributions along the wingspan for angles of attack (α) 0°, 4°, and 8°. For clarity, plot the results in three figures, one for each *AoA*, with five curves in each (corresponding to the five aspect ratios).
  - Plots of induced C_L and C_D as a function of angles of attack, with *AoA* running from -4° to 8°.
- *Question 3:* Plots of $\tilde{\Gamma}$, α_i, and the corresponding induced C_L and C_D, as functions $\tilde{x}$.
- *Question 4:* Plots of $\tilde{\Gamma}$, α_i, and the corresponding induced C_L and C_D, as functions $\tilde{x}$. Show the results on four figures (one for each variable) with five plots on each (one for each tip angle).